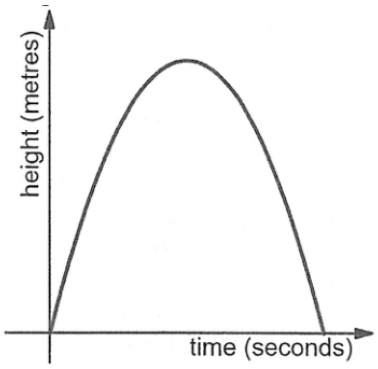
Quadratic Modelling

Questions 1–10

Question 1. A ball is thrown vertically upwards with a speed of 19.6ms^{-1}. The height of the ball, h metres, may be modelled by the equation $h = 19.6t - 4.9t^2$ where t is the time given in seconds.

a) Find the height of the ball after 2 seconds.
b) Find the times at which the ball is at a height of 14.7 metres above the ground.
c) Explain what your two solutions in the previous part represent.

Question 2. A ball is hit into the air, the graph below shows its vertical height above the ground before it hits the ground.



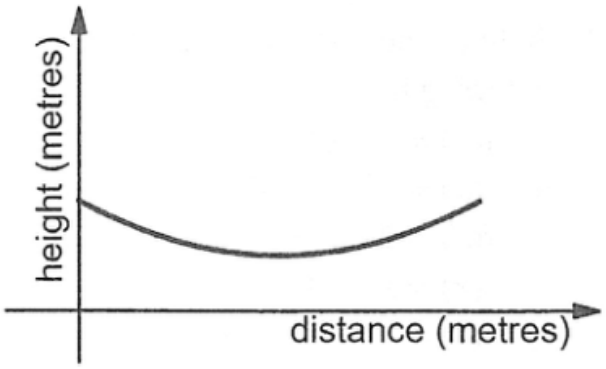
A model for the height of the ball, h metres at time t seconds is given by $h(t) = 10t - 2t^2$. Find

a) the time when the ball hits the ground.
b) $h(t)$ in the form $k(t-p)^2 + q$ where p, q and k are constants to be found.
c) the maximum height the ball reaches and the time at which it reaches the maximum height.

An improved model starts the ball at $h = 1$.

d) State the new model in the form $h(t) = at^2 + bt + c$.
e) State the coordinate of the maximum height of the curve of this new model.

Question 3. Two identical fence posts are 2.5 metres apart and a rope is hung between them. The curve $y = \frac{1}{100}x^2 - \frac{1}{40}x + \frac{1}{2}$ may be used to model the rope, where y is the vertical distance in metres above the ground and x is the horizontal distance in metres from the left post. The curve is shown in the diagram below:



a) State the height of the posts.
b) Rearrange the model into the form $a(x-b)^2 + c$ where a, b and c are constants to be found.
c) State the horizontal and vertical position of the minimum height of the rope.
d) State why the model is not effective for $x > 2.5$.

Question 4. The path of a swimmer may be modelled by $d = 0.1x^2 - x + 0.9$ for $x \geq 0$ where d is the depth of the swimmer in relation to the water surface and x metres is the distance from the poolside.

a) Sketch the graph of the model.

According to the model:

b) Find the depth that the swimmer swims below the surface of the water.
c) Determine the horizontal distance the swimmer swam under the water.
d) State how far from the pool edge the swimmer is when they first contact the water.
e) From what value of x does the function become an unrealistic model for the motion of the swimmer?

Question 5. The income £I made by a company on various cleaning services they provide may be modelled by the equation $I = \frac{5}{2}x(40 - x)$ where £x is the price charged for each individual service.

a) Sketch the model for $0 \leq x \leq 40$.

b) Rearrange the model into the form $I = p - \frac{5}{2}(x - q)^2$ where p and q are constants to be found.

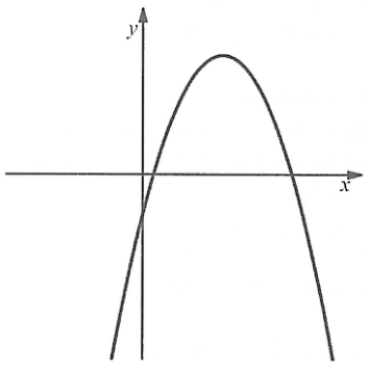
According to the model:

c) Find the maximum income that is made.

d) Find the price of the service that gives the maximum income.

e) Find the prices which give an income of £600.

Question 6. The profit of a business selling books depends on the price at which each book is sold. The equation $y = \frac{15}{2}x - \frac{25}{2} - \frac{5}{8}x^2$ is used to model the relationship between the profit and the price where £x is the selling price of each book and y pence is the profit achieved at that price. A sketch of the curve is shown in the diagram below:



Find

a) the selling price that generates the maximum profit.

b) the value of the maximum profit.

c) the selling price that gives no profit.

d) the loss if the books are given away.

Question 7. The growth of two flowers may be modelled by quadratic curves as follow:
Flower A: height y cm after x days is given by $y = -\frac{2}{45}x^2 + \frac{8}{3}x$.
Flower B: height y cm after x days is given by $y = -\frac{1}{10}x^2 + 3x + \frac{15}{2}$.

a) Calculate the number of days which have passed when they are the same height.
b) Find the maximum height of the curve representing flower B.
c) Explain why the growth of the flowers may not best be modelled by these models.

Question 8. A ball is projected from the ground. It reaches a maximum height of 10 metres at 2 seconds since being projected. Find a model for the trajectory of the ball in the form $h = at^2 + bt$ where h is height (metres), t is time (seconds) and a and b are constants to be found.

Question 9. A stone is projected from the ground. It starts at a horizontal position of a metres from a wall. It is modelled by $y = -\frac{1}{8}x^2 + \frac{11}{8}x - \frac{57}{32}$ where x and y are the horizontal and vertical distance in metres respectively from the bottom of the wall.

a) Find the value of a.
b) Find the total horizontal distance travelled by the ball before it hits the ground.
c) Find the maximum height that the ball reaches.
d) A tree of height 1.2 m is 7.5 m from the first wall, does the stone hit the tree?

Question 10. An arch has a span of 350m and a maximum vertical height of 140m above ground level. Let x be the horizontal distance from the left-most point of the arch in metres and y be the vertical distance above the ground. Find a model in the form $y = \frac{p}{q}x(r - x)$ where p, q and r are integers to be found.